

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SHORT ANSWER TEST

Course Code:	CSA16	Course Title:	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	10% of CIA	Total Marks:	100 (5 Questions × 20 Marks)
CO1 Statement:	<i>Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2</i>		
Student Name:	B. V praveenika	Reg. No.:	192472200
Section / Batch:		Date:	15-07-2026

Instructions to Students

- Answer all questions. Each question carries 20 Marks. Total: 100 Marks.
- Clearly show all requirements, process, operations.
- Use appropriate models, schema represented scenario wise
- Justify each step in different dimension specified and measures clearly.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Short Answer Test	Architecture of Data Warehouse, ETL, OLAP	Q1 – Q5	Q1 –20 Q2 –20 Q3 –20 Q4 –20 Q5 –20
GRAND TOTAL:		100 Marks	

SECTION A – SHORT ANSWER TEST

1. Suppose that a data warehouse for City Hospital consists of the following four dimensions: patient, doctor, department, and time, and two measures visit_count and avg_charge. At the lowest conceptual level, avg_charge stores the actual consultation fee charged for a patient visit. At higher conceptual levels, it stores the average fee.

(a) Draw a snowflake schema diagram for the data warehouse.

(b) Starting with the base cuboid [patient, doctor, department, time], what specific OLAP operations should be performed to list the average consultation fee department-wise for each year?

(c) If each dimension has four levels (including all), how many cuboids will this cube contain (including base and apex cuboids)?

2. A smart city project is collecting data from multiple sources, including:

- Traffic sensors for vehicle counts and congestion levels.
- IoT devices for monitoring air quality, noise levels, and energy consumption.
- Public transportation systems for ridership data and punctuality.
- Citizen feedback from a mobile app.

Questions:

Fact and Dimension Design:

- Design a star schema to analyze traffic congestion patterns, energy consumption trends, and public transportation efficiency.
- What measures and dimensions would you include? Justify your design.

3. Suppose that a banking data warehouse contains two fact tables: Transaction_Fact and Loan_Fact, sharing dimensions customer, branch, time, and account_type.

- Transaction_Fact → transaction_count, amount
- Loan_Fact → loan_amount, interest

(a) Draw a galaxy schema diagram.

(b) Identify OLAP operations to obtain branch-wise average transaction amount and loan amount yearly.

(c) If each dimension has four levels, calculate total cuboids.

4. A manufacturing company monitors product quality and production efficiency. The data warehouse contains:

- Dimensions:
 - Product: Category, type, and batch.
 - Factory: Location, department, and line number.
 - Time: Year, quarter, month, and week.
 - Supplier: Region, rating, and contract type.
- Measures:
 - Defect rate.
 - Production units.
 - Supplier reliability percentage.

Questions:

- Roll-up: Summarize defect rates by product category for the current year across all factories.
- Drill-down: Focus on a specific factory to analyze defect rates by department and product type.
- Slice: Isolate production unit data for suppliers with a rating of 4+ across all regions.
- Dice: Analyze defect rates and supplier reliability for selected product categories and factory locations.

5. A retail chain manages inventory and sales data across multiple stores. The data warehouse contains:

- Dimensions:
 - Product: Category, subcategory, and SKU.
 - Store: Region, city, and store ID.
 - Time: Year, quarter, month, and day.
 - Customer: Membership tier, age group, and gender.
- Measures:
 - Inventory turnover.
 - Total sales revenue.
 - Units sold.

Questions:

1. Roll-up: Summarize inventory turnover by product category across all stores for the last quarter.
2. Drill-down: Focus on sales data for electronics in a specific region, analysing by subcategory and SKU.
3. Slice: Isolate sales revenue data for male customers in the Bronze membership tier across all stores.
4. Dice: Analyze sales revenue and units sold for selected cities and product categories during specific time periods.

Code of Conduct

(192472200)

I, B. V. pravaulika (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B. V. pravaulika

RUBRICS FOR CALCULATION

Criteria	Wt	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Marks Evaluation:

Question No.	Understanding of Concepts (25%)	Explanation (25%)	Application (20%)	Organization (15%)	Accuracy (10%)	Clarity (5%)	Total Marks
Q1 (20 Marks)							
Q2 (20 Marks)							
Q3 (20 Marks)							
Q4 (20 Marks)							
Q5 (20 Marks)							
Overall Total (100)							

Faculty in-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

city Hospital data warehouse

(a) Snowflake Schema Design

- Fact - Hospital : contains foreign keys (patient-id, doctor-id, time-id, dept-id) and measures (visit-count, avg-charge).
- Dim - patient : patient-id, name, age, gender.
- Dim - Doctor : doctor-id, name, specialization, dept-id.
- Dim - Department : dept-id, dept-name, head-of-dept.
- Dim - Time : time-id, day, month-id.
- Dim - Month : month-id, month-name, year.

(b) OLAP operations

- Roll-up : Roll up on the time dimension from the day / visit level to the year level.
- Drop : Drop the patient and doctor dimension from the cuboid view.
- Group By : Aggregate the data by department and year.

(c) cuboid count calculation

- Formula : $L_1 \times L_2 \times L_3 \times L_4$
- calculation : $4 \times 4 \times 4 \times 4 = 256$ cuboids.

2. Smart City Project

• Dimensions and Measures

• Dimensions

- Dim - time (hour, Day, month, year)
- Dim - Location (street, Neighborhood, District)
- Dim - Source (sensor type, system Name)

• Measures

vehicle count, logestion-level, energy-consumption, ridership-count, punctuality-rate, feedback-score.

• Schema Design & Justification

- Design: A centralized Fact-smartcity table linked directly to the independent dimensions listed above in a star-schema pattern.

• Justification:

Simplicity and performance reduces complex table joins, allowing rapid querying across massive datasets of city sensor logs.

3. Banking Data warehouse

(a) Galaxy schema diagram

customer-Dim Branch-Dim Time-Dim Account-Type-Dim

Transaction-Fact	Loan-Fact	Transaction	Loan-Fact
- customer-id	- customer-id	- fact	- loan-
- branch-id	- branch-id	- transaction	- amount
- time-id	- time-id	- count	- interest
- Account-id	- Account-id	- Amount	

operations

- Roll-up on branch to group by branch location →
- Roll-up on time to aggregate data to the year level →
- Drop customer and account-type dimensions, for transaction.
- Roll-up on branch to group by branch location → Roll-up on time to aggregate data to the year level → Drop customer and account-type dimensions.

(c) Total cuboids calculation.

The system contains 4 shared dimensions, each having 4 levels.

Calculation per Fact table: $5^4 = 625$ cuboids

Total for Galaxy schema: Since both fact tables share the exact same 4 dimensions, the maximum collection of unique cuboids is 625.

4 OLAP Queries Execution Plan.

- Roll-up: Group the Defect rate measure by product category while filtering Time. year = current year. Drop the Factory dimensions levels to summarize across all physical plants.
- Drill-down: Apply a hard filter on Factory. Location = [specific Factory Name]. Move down the hierarchy of the product dimension: product category → product. Type →

product, Batch to evaluate specific defect distribution.

3. slice : Apply a fixed predicate constraint: Supplier.Rating ≥ 4 . Extract the sub-cube for the measure production units plotted across all available supplier, Region coordinates.
4. Dice: Apply multiple specific dimension filters simultaneously.

1. product.category IN ([selected categories])

2. Factory.Location. IN ([selected Locations])

3. Analyze the remaining metrics cell intersections for defect rate and supplier reliability percentage.

5. OLAP Queries Execution plans.

1. **Roll-up**: Filter Time.Quarter = [Last Quarter]. Group the Inventory turnover measure by product.category. Remove individual store restrictions to aggregate results across all store locations globally.

2. **Drill-down**: Filter product.category = "Electronics" and Filter by the target geographical region, step down the product hierarchy from category $\rightarrow$ subcategory $\rightarrow$ sku to view granular revenue breakdowns.

3. **slice**: slice the data cube by setting two static predicates: customer.Gender = Male AND customer.

Membership_tier = 'Bronze'. project the metric total sales revenue across all remaining store location.

Dice : Apply strict multi-attribute filters across dimensions.

1. store.city IN ([selected cities])
2. product.category IN ([selected categories])
3. Time.Quarter / Time.month IN ([selected time periods])
4. Analyse the values in the interesting cells for both total sales revenue and units sold.

